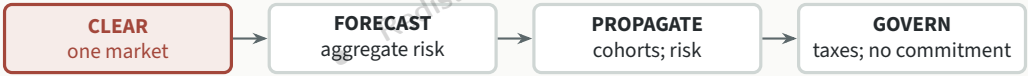
Session 4.1

Aiyagari, Solved Twice: A Fixed Point, Then a Law of Motion

The classical solve returns one stationary distribution. An actor–critic solve with the distribution as its state returns how every distribution evolves

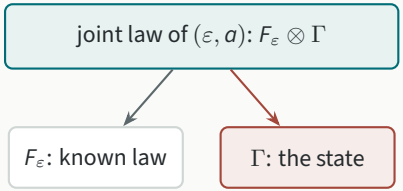


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The Economy



ε is redrawn every period, independently of a, so today's assets tell nothing about today's ε.

- $\log \varepsilon \sim N(0, 0.2^2)$, i.i.d.; $c + a' = (1+r)a + w\varepsilon l$, $a' \geq 0$.
- Utility $u(y) = y^{1-\sigma}/(1-\sigma)$ of the bundle $y = c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}$: no wealth effect on hours, so $l(\varepsilon) = (w\varepsilon/\psi_l)^{1/\theta_l}$.
- $Y = K^\alpha L^{1-\alpha}$; r (net of δ) and w from marginal products; $K = \int a \, d\Gamma$.
- $\Gamma_{t+1} = \Phi(\Gamma_t)$. With i.i.d. ε the distribution of assets, Γ, is the whole state.
- **Prices depend on Γ, so the saving rule is $g(\varepsilon, a; \Gamma)$.**

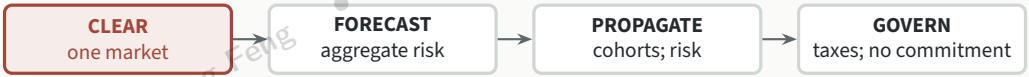
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\beta = 0.95, \sigma = 2, \alpha = 0.34, \delta = 0.06, \theta_l = 0.32, \psi_l = 1.5$.

Session 4.1

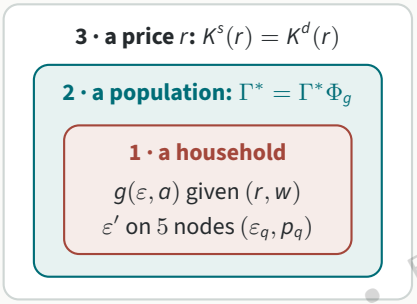
Stage 1A: The Classical Solve

A household, a population and a price: three fixed points, one inside the other



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Three Problems, Nested



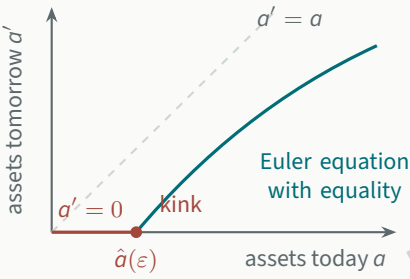
one evaluation of the outer residual: a full household solve and a full distribution solve

- **A household.** Given (r, w): $V(\varepsilon, a) = \max_{a' \geq 0} u(y) + \beta \sum_q p_q V(\varepsilon_q, a')$; with i.i.d. ε the expectation depends on a' alone.
- **A population.** The rule moves any Γ to $\Gamma' = \Gamma\Phi_g$; find the invariant $\Gamma^* = \Gamma^*\Phi_g$.
- **A price.** Find r at which $K^s(r) = \int a \, d\Gamma_r^*$ equals the firms' $K^d(r)$; w follows from r.
- $D(r) = K^d(r) - K^s(r)$: both terms must depend on r, or $D \equiv 0$, invisibly.
- **Each layer is a fixed point and contains the next: one evaluation of D costs a full dynamic program.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Aiyagari, *QJE* (1994).

The Borrowing Limit Puts a Kink in the Rule



- With $a' \geq 0$: $u'(y) \geq \beta(1+r)\,\mathbb{E}\,u'(y')$, with equality when $a' > 0$.
- Below $\hat{a}(\varepsilon)$ the household would borrow; it cannot, so $a' = 0$: a kink in the rule.
- A kink defeats methods that assume smoothness; on a grid it is one comparison per node.
- Here 0.009% sit exactly at $a = 0$: i.i.d. risk needs only a small buffer. Persistence changes that (last section).
- **The constraint is handled by a comparison, not a solver.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A.

The Household: the Endogenous Grid Method

- Grid on *tomorrow's* assets, Euler equation inverted, no search. In the bundle the budget is linear: $y + a' = (1+r)a + \phi(\varepsilon)$, with $\phi(\varepsilon) = w\varepsilon l(\varepsilon) - \psi_l \frac{l(\varepsilon)^{1+\theta_l}}{1+\theta_l}$, earnings net of the hours' disutility.

One sweep from the current guess g, on the fixed grid $a'_1 < \cdots < a'_n$

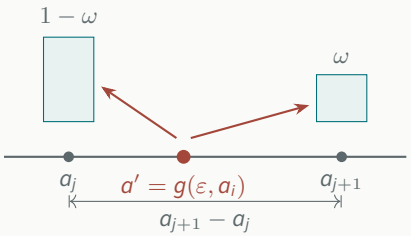
1. tomorrow's bundle at every (a'_j, ε_q): $y'_{jq} = (1+r)a'_j + \phi(\varepsilon_q) - g(\varepsilon_q, a'_j)$, no interpolation
2. $E_j = \sum_q p_q\, u'(y'_{jq})$: one number per a'_j, since ε' is independent of today
3. invert the Euler equation: $\tilde{y}_j = \left[\beta(1+r)E_j\right]^{-1/\sigma}$
4. today's assets that lead here, one per ε: $\tilde{a}_j(\varepsilon) = \left[\tilde{y}_j + a'_j - \phi(\varepsilon)\right]/(1+r)$
5. interpolate $(\tilde{a}_j(\varepsilon), a'_j)$ onto the fixed grid: the new $g(\varepsilon, a)$; $a' = 0$ for $a < \tilde{a}_1(\varepsilon)$

- No root-finding, no maximisation. Repeat until $\max|g_{\text{new}} - g| < 10^{-10}$; 1500 nodes, dense near the limit.
- **The constraint is where the interpolation runs out: a comparison, not a solver.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Carroll, *Economics Letters* (2006).

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The Young Lottery



- The rule sends (ε, a_i) to $a' = g(\varepsilon, a_i)$, between two nodes: where does its mass go?
- Split it: $\omega = \frac{a'-a_j}{a_{j+1}-a_j}$ to a_{j+1}, $1-\omega$ to a_j. Rounding to one node would bias every moment.
- Then ε' is drawn with weights p_q. Mass is conserved and the conditional mean of a' is exact: Young (2010), often unnamed as "linear interpolation".
- **Deterministic. A simulated panel adds sampling noise to what we will later differentiate.**

Source: Young, *JEDC* (2010).

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From a Rule to a Population

- Lottery weights and ε draws define the **transition matrix** Φ_g (two nonzeros per (i, q)):

$$\Gamma' = \Gamma\,\Phi_g, \qquad (\Phi_g)_{ij} = \textstyle\sum_q p_q\,\omega_{iq\to j}, \qquad \omega_{iq\to j} = \text{share of } (\varepsilon_q, a_i)\text{'s mass sent to } j.$$

Three ways to $\Gamma^* = \Gamma^*\Phi_g$	**What it costs**
iterate $\Gamma_{t+1} = \Gamma_t\Phi_g$	simple; slow near a unit root
dense solve: $\Gamma^*(\Phi_g - I) = 0, \sum_i \Gamma^*_i = 1$ replacing one row	$O(n^3)$, exact; **Stage 1A**, $n = 1500$
the sparse eigenvector of $\Phi_g^\top$ at eigenvalue 1	fastest at scale

- Check the top of the grid: its top tenth holds 1.2×10^{-4} of the mass here; at 10^{-2} the grid would set K.
- **A histogram plus the lottery is the law of motion, as a matrix; Stage 1B rebuilds it every date.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Young (2010).

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The Dense Solve: Trade One Redundant Equation for $\sum_i \Gamma_i^* = 1$

A toy chain, $n = 3$:

$$\Phi_g = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$(\Phi_g^\top - I)\,\Gamma^{*\top} = 0$, last row swapped:

$$\begin{bmatrix} -\frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \Gamma_1^* \\ \Gamma_2^* \\ \Gamma_3^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

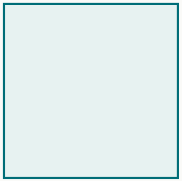
$$\Rightarrow \Gamma^* = \left(\tfrac{1}{4},\ \tfrac{1}{2},\ \tfrac{1}{4}\right)$$

- $(\Phi_g^\top - I)\,\Gamma^{*\top} = 0$: n equations, n unknowns.
- Rows of Φ_g sum to one, so the equations add up to $0 = 0$: one is redundant, and any multiple of Γ^* solves them.
- Swap it for $\sum_i \Gamma_i^* = 1$ (red): invertible, scale fixed.
- One LU factorisation: $\frac{2}{3}n^3$ operations on all n^2 entries, zeros included; $n^2 = 2{,}250{,}000$ here.
- **Exact in one shot; the cost grows like n^3 however sparse Φ_g is.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A.

The Eigenvector Route: Multiply by $\Phi_g^\top$, Never Factor It



dense LU works on all n^2 entries



$\Phi_g^\top v$ touches at most $2Qn$

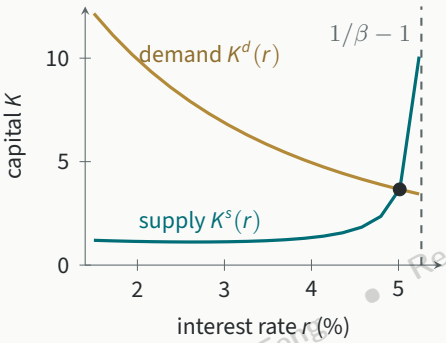
here $14{,}534$ against $2{,}250{,}000$

- Same equation, read as $\Phi_g^\top v = 1 \cdot v$: 1 is the largest eigenvalue; normalise after, $\Gamma^* = v / \sum_i v_i$.
- Plain iteration is the power method: error $\propto |\lambda_2|^t$. Toy: $\lambda_2 = \frac{1}{2}$. Here $\lambda_2 = 0.9972$: $t = \ln 10^{-10} / \ln \lambda_2 \approx 8{,}200$.
- Arnoldi uses the same products $v, \Phi_g^\top v, (\Phi_g^\top)^2 v, \ldots$ but combines them all: far fewer.
- **Same Γ^*; the cost is products over $2Qn$ nonzeros, so it wins as n grows.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A transition matrix; λ_2 by Arnoldi (ARPACK).

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The Outer Loop: Bisection in a Proven Bracket, Not Newton



- The bracket is known: demand falls in r, supply diverges as $r \to 1/\beta - 1$. Prove the sign change, then halve to 10^{-8}.
- Each evaluation is one warm-started EGM and one distribution solve. Not Newton: supply is only piecewise smooth in r (the binding set jumps), so a finite difference is no derivative.
- $r^* = 5.014\%$ against $1/\beta - 1 = 5.263\%$: precautionary saving. $K^* = 3.6595$, $K/Y = 3.09$; 30 CPU-seconds in all.
- $\beta(1 + r^*) = 0.99763$**: so close to one, and it returns in every section that follows.**

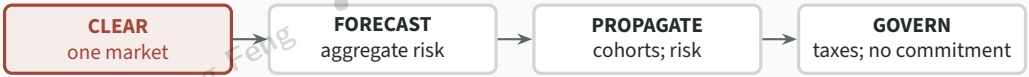
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Aiyagari, *QJE* (1994).

Session 4.1

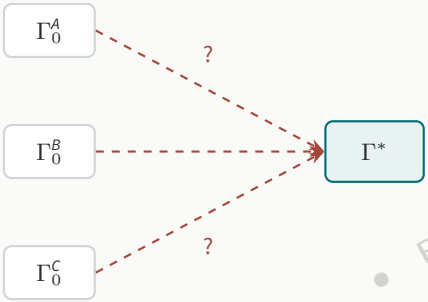
What the Classical Solve Cannot Answer

One fixed point is not a law of motion



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One Fixed Point, Not a Law of Motion



each dashed path is a separate problem: a sequence of prices, solved backward and forward

- Stage 1A says where the economy rests, not how it gets there.
- A transition from one Γ_0 is a price path and a shooting loop (Deck 4.3). A second Γ_0 is a second loop.
- Welfare along the way, or a policy that reacts to Γ, needs the answer at every Γ.
- **The unknown changes from a number, r^*, to a function of a distribution: $g(\varepsilon, a; \Gamma)$.**

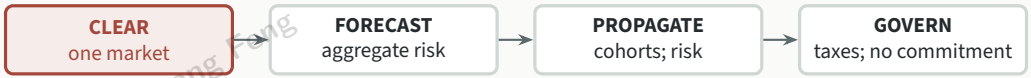
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Deck 4.3, *The Unknown Is a Path of Prices*.

Session 4.1

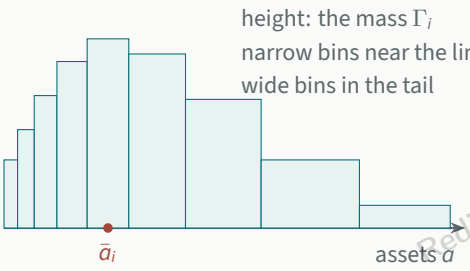
Characterising the Evolution of Γ

A histogram, a quadrature, and a transition matrix rebuilt inside the computational graph



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A Histogram and a Quadrature

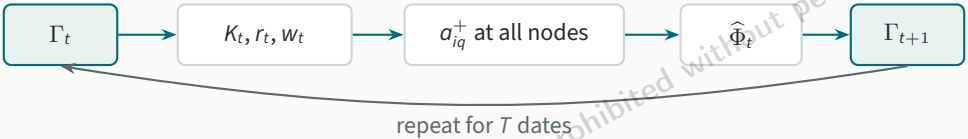


- $N = 140$ bins on $[0, 16]$ with midpoints $\bar{a}_i$; the state is $\Gamma = (\Gamma_1, \ldots, \Gamma_N)$, $\sum_i \Gamma_i = 1$, $K = \sum_i \Gamma_i \bar{a}_i$.
- ε: $Q = 5$ Gauss–Hermite nodes ε_q, weights p_q, as in Stage 1A. An expectation is a five-term sum.
- Grid chosen once from the benchmark: no bin above 1.6% of households, nothing off the top.
- **The discretisation is part of the model. Fix it for every method compared.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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The Law of Motion, Rebuilt at Every Date



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From one distribution Γ_0, for $t = 0, \ldots, T-1$

1. $K_t = \sum_i \Gamma_{t,i} \bar{a}_i$; r_t, w_t from the firm, hours in closed form: no price loop
2. saving $a_{iq}^+ = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma_t)$ at every node; consumption from the budget
3. build the $N \times N$ matrix $\widehat{\Phi}_t$ and set $\Gamma_{t+1} = \Gamma_t \widehat{\Phi}_t$

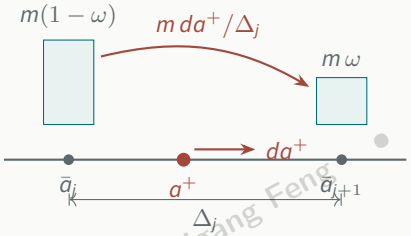
- **$\widehat{\Phi}_t$ depends on Γ_t and on the network, so it is rebuilt at every date of every rollout: computed, never simulated, inside the graph.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Building $\widehat{\Phi}$, and Why This Kernel

Stage 1A's lottery, now from the network's a^+: for every source midpoint $\bar{a}_i$ and node ε_q

1. evaluate the saving network once: $a^+ = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma)$
2. find the bracket $\bar{a}_j \le a^+ \le \bar{a}_{j+1}$, and the weight $\omega = (a^+ - \bar{a}_j)/(\bar{a}_{j+1} - \bar{a}_j)$
3. add $p_q(1-\omega)$ to $\widehat{\Phi}_{ij}$ and $p_q\,\omega$ to $\widehat{\Phi}_{i,j+1}$



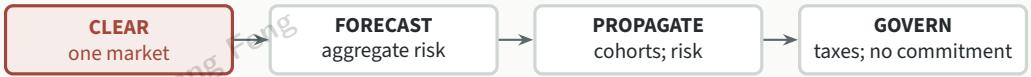
- **Exact in the mean**: the only split between neighbours with mean a^+.
- **Mass by construction**: each row of $\widehat{\Phi}$ sums to one.
- **Differentiable**: da^+ moves mass $m\,da^+/\Delta_j$ between neighbours, and autodiff chains it across dates.
- **The same lottery as Stage 1A; its derivative is what the rollout's gradient needs.**

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Session 4.1

Stage 1B: An Actor–Critic Solve

The saving rule is the actor, the household's value is the critic, and both read the histogram



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The Problem the Two Networks Solve

- At a distribution Γ, each household solves, taking Γ, prices and Γ_+ as given,

$$V(\varepsilon, a; \Gamma) = \max_{0 \le a' \le \bar{a}} \Big\{ u(y) + \beta \sum_{q'} p_{q'} \, V(\varepsilon_{q'}, a'; \Gamma_+) \Big\}, \qquad y = c - \psi_l \tfrac{l^{1+\theta_l}}{1+\theta_l},$$

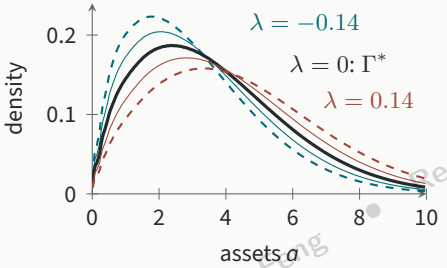
  with $c = (1 + r(\Gamma))\, a + w(\Gamma)\, \varepsilon\, l(\varepsilon; \Gamma) - a'$ and $\Gamma_+ = \Gamma\, \widehat{\Phi}(\Gamma; \theta)$.
- **Actor**: $a' = g_\theta(\varepsilon, a; \Gamma) = \max\{\tilde{a}_\theta, 0\}$, capped at $\bar{a}$: the **borrowing limit** is built into the rule, so $a' \ge 0$ holds at every node; where it binds, $c = (1 + r)a + w\varepsilon l$.
- **Critic**: $V_\psi \approx V$. Its Bellman target is computed from a trailing copy ψ^- of its weights, nudged after each critic step ($\psi^- \leftarrow 0.98\, \psi^- + 0.02\, \psi$), so the target holds still while ψ learns: the **target network**.
- **Both read the whole histogram: two hidden layers of 256 units on (ε, a) and the 140 masses.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

Preparing the Batch: Whole Distributions Around Γ^*

Γ^* from Stage 1A → draw $\lambda \sim U[-0.14, 0.14]$ → $\Gamma_i^e \propto \Gamma_i^* \, e^{\lambda \bar{a}_i}$, renormalise → a batch of 128 Γ^e



- A training example is a whole distribution Γ^e, never a household; 128 fresh ones per iteration.
- Tilt Γ^* (five shown, $\lambda = 0, \pm 0.07, \pm 0.14$): $\lambda > 0$ moves mass to the rich, $\lambda < 0$ to the poor; K from 3.04 to 4.45.
- Each node $(\bar{a}_i, \varepsilon_q)$ of each Γ^e enters the loss: $N \times Q$ states per aggregate state.
- **Γ^e is today's distribution, $\Gamma_+^e = \Gamma^e \widehat{\Phi}(\Gamma^e)$ its successor; Γ^* is only the band's centre.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

The Critic Step: Fit the Bellman Equation

- At every node $(\bar{a}_i, \varepsilon_q)$ of every Γ^e, with $a^+_{iq} = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma^e)$ from the current actor, minimise over ψ

$$\mathcal{L}_V(\psi) = \underset{e,i,q}{\text{mean}} \left[V_\psi(\varepsilon_q, \bar{a}_i; \Gamma^e) - b^e_{iq} \right]^2,$$
$$b^e_{iq} = U^e_{iq} + \beta \sum_{q'} p_{q'} \, V_{\psi^-}\left(\varepsilon_{q'}, a^+_{iq}; \Gamma^e_+\right).$$

- U^e_{iq}: flow utility at a^+_{iq}. b is held fixed while ψ moves; its ε'-sum is exact, so no sampling noise.
- **One step: one Adam update of ψ on the whole batch (rate 3×10^{-3}), then $\psi^- \leftarrow (1 - 0.02)\, \psi^- + 0.02\, \psi$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

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The Actor Step: Maximise the Bellman Right-Hand Side

- Critic fixed, the actor maximises the Bellman right-hand side at every node of every Γ^e, with $a^+_{iq} = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma^e)$:

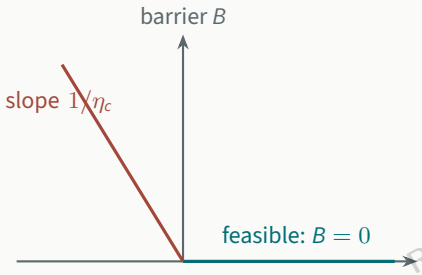
$$J(\theta) = \text{mean}_{e,i,q}\left[u(y^e_{iq}) + \beta\, C(a^+_{iq})\right], \qquad C(a') = \textstyle\sum_{q'} p_{q'}\, V_{\psi^-}(\varepsilon_{q'}, a'; \Gamma^e_+).$$

- One step: one Adam update of θ along $\nabla_\theta J = \text{mean}_{e,i,q}\left[\partial_{a^+}(u + \beta C)\right] \partial_\theta a^+_{iq}$, rate 5×10^{-5}.
- C's *value* is the critic's, its *slope* the envelope condition; so at an interior node $\partial_{a^+}(u + \beta C) = -u'(y) + \beta \sum_{q'} p_{q'}(1 + r_+)\, u'(y_{+,q'})$, the Euler residual, driven to zero.
- **At the limit the clamp has zero slope: no Euler equation at the constraint, none needed without a government; that household contributes nothing.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. $y_{+,q'}$: the bundle the current rule leaves at $(\varepsilon_{q'}, a'; \Gamma_+)$.

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Why the Barrier, and Why the Slope Is Replaced



- A proposed a' can make c or y negative, where $u(y)$ is undefined.
- $B = \left[\max(-c, 0) + \max(-y, 0)\right]/\eta_c$, $\eta_c = 10^{-4}$: a hinge, zero wherever the proposal is feasible.
- The actor maximises $J - \kappa B$, $\kappa = 1$; at the optimum $B = 0$.
- Why the envelope slope: at $\beta R = 0.99763$ the Euler terms nearly cancel, and a fitted V's slope is too noisy to resolve that.
- **One barrier for feasibility, one exact slope; neither changes what the actor maximises.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

What Pins K: the Rollout Penalty

$$\mathcal{L}_g(\theta) = -J(\theta) + \kappa \operatorname*{mean}_{e,i,q} B^e_{iq}(\theta) + \lambda_K \frac{1}{T}\sum_{t=1}^{T}\left(K_t(\theta) - K^*\right)^2$$

- $-J$ is the actor's Bellman objective; κ mean B is the feasibility barrier of the last frame, $\kappa = 1$: zero wherever consumption is feasible, so it never moves the optimum.
- The **rollout penalty**: roll Γ^* forward $T = 100$ periods under the current rule, inside the graph ($\Gamma_{t+1} = \Gamma_t \widehat{\Phi}(\Gamma_t; \theta)$, $K_t = \sum_i \Gamma_{t,i}\bar{a}_i$), and penalise the capital path's squared distance from K^*, weight $\lambda_K = 1000$.
- Why: with $\beta R = 0.99763$ a one-signed bias in saving is amplified about $1/(1-\beta R) \approx 420$ times in the stationary K. A one-step residual cannot see it; a hundred dates can.
- 2 **Adam steps on $\mathcal{L}_g$ per iteration. The rollout, not a residual, pins K.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

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The Algorithm

Stage 1B: the household's actor–critic, no government

1 initialise the actor by regression on EGM rules at constant prices, one per sampled K
2 pre-fit the critic to that actor, 6000 steps, so the actor cannot exploit a bad critic
3 draw 128 tilted distributions Γ^e
4 $\quad 8$ critic steps: each one Adam update of ψ on $\mathcal{L}_V$, then $\psi^- \leftarrow (1-0.02)\psi^- + 0.02\,\psi$
5 $\quad 2$ actor steps: each one Adam update of θ on $\mathcal{L}_g$
6 repeat 4000 times; at intervals, find the actor's own stationary Γ and score it

- A **step**: one gradient update of one network on the batch of 128 distributions.
- 8 critic steps for 2 actor steps: the critic's target moves whenever the rule moves, so it takes more and larger steps to stay fitted.
- **The actor's slope comes from the envelope condition; what the critic steps buy is a fitted $V_\psi(\varepsilon, a; \Gamma)$, the household value Deck 4.4 loads.**

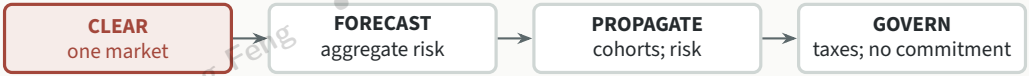
Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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Session 4.1

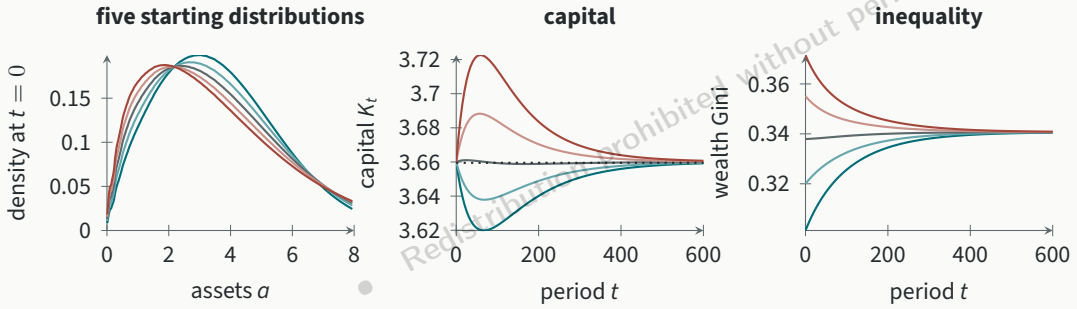
What It Returns, and Where It Stops

The distribution in motion, and why persistence changes the problem



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Same Capital, Different Inequality

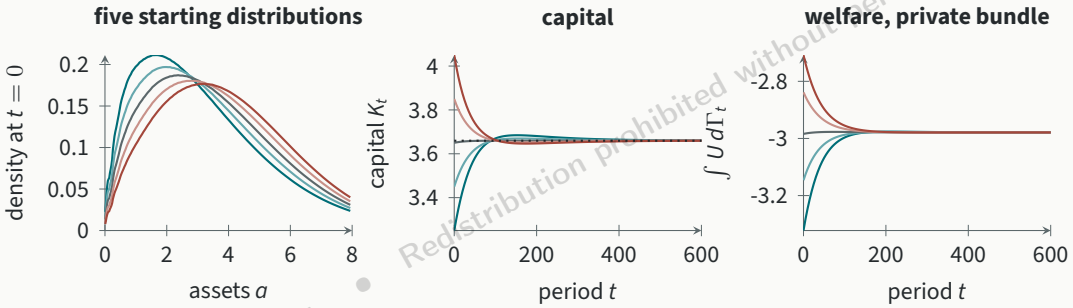


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- All start at $K_0 = 3.660$, Gini 0.301–0.372; all reach Gini 0.341, from below and above.
- K **leaves and returns (**3.620–3.723**). A state summarised by its mean would see nothing.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^*.

Same Spread, Different Capital



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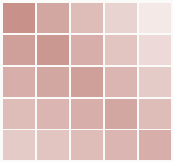
- K_0 from 3.25 to 4.05: K returns in about a hundred periods, the Gini far later.
- **Welfare along the path, from any start: what Stage 1A cannot return.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^*. No government, so no public good.

Where the Characterisation Gets Hard



N masses



$N \times n_\varepsilon$ masses

a and ε become correlated: the poor are poor because their ε was low

- Persistent ε: the joint law of (ε, a) no longer factorises. The state is a joint histogram, and $\widehat{\Phi}$ must carry the chain Π.
- With i.i.d. risk only 0.009% sit at the limit. In the lab's economy, with persistence 0.9, 2.9% do: runs of low ε walk them down to it.
- An aggregate shock z makes the map $\Phi(\Gamma, z, z')$ random, to be learned on every branch.
- **Each step multiplies the size of $\widehat{\Phi}$ and of every rollout in memory.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the lab, block A; Krusell & Smith, *JPE* (1998).

Bridge: When Φ Cannot Be Written Down, Simulate It

This deck. Γ is a histogram, Φ a matrix computed in the graph: exact, deterministic, differentiable.

persistence, aggregate risk

Next. Γ is a simulated panel, and the network learns from the states that the panel visits.

- Deep Equilibrium Nets and DeepHAM simulate households, and train where the simulation goes.
- The price: sampling noise, no derivative through a head count, coverage only where the policy goes.
- The gain: persistence and aggregate risk cost a longer simulation, not a bigger matrix.
- **Deck 4.2: three such methods. Deck 4.4: this histogram, with a government.**

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Source: Azinovic, Gaegauf & Scheidegger, *IER* (2022); Han, Yang & E (2026).